

Supplementary Documents for "Epileptic Seizure Detection Using CHB-MIT Dataset: The Overlooked Perspectives"

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Appendix-A

All Features Details

Given a time-series signal $x = \{x_i : i = 1, 2, 3, \dots, n\} = \{x_1, x_2, x_3, \dots, x_n\}$, with n being the number of data points. The following features from different domains were extracted and used for seizure classification. For the mathematical notations, all the symbols are kept consistent for the same functions or variables for the entire paper, starting from top to bottom.

Time-domain Features

Several linear and non-linear features were extracted from the time domain of the signal x that explain the periodicity and related information about the signal.

Linear statistical measures from the signal can be estimated as follows:

Maximum (*Max*): The highest positive value or data point of the signal can be calculated as-

$$Max(x) = \max(\{x_i : i = 1 \dots n\}) \quad (0.1)$$

Minimum (*Min*): The highest negative value or data point of the signal can be calculated as-

$$Min(x) = \min(\{x_i : i = 1 \dots n\}) \quad (0.2)$$

Mean (*Mean*): The average value of all data points of the signal can be calculated as-

$$Mean(x) = \mu = \frac{1}{n} \sum_{i=1}^n x_i \quad (0.3)$$

Median (*Med*): The Median value of all data points of the signal can be calculated as-

$$Med(x) = \begin{cases} x_{(\frac{n+1}{2})} & \text{if } (n\%2) \neq 0 \\ \frac{1}{2}(x_{(\frac{n}{2})} + x_{(\frac{n}{2}+1)}) & \text{if } (n\%2) = 0 \end{cases} \quad (0.4)$$

Standard Deviation (*Std*): The Standard Deviation of all data points of the signal from its mean value can be calculated as-

$$Std(x) = \sigma = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2} \quad (0.5)$$

Variance (*Var*): The Variance of the data points of the signal can be calculated as-

$$Var(x) = \sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2 \quad (0.6)$$

Kurtosis (*Kur*): The Kurtosis or steepness of all data points of the signal can be calculated as-

$$Kur(x) = \frac{\sum_{i=1}^n (x_i - \mu)^4}{\sigma^4} \quad (0.7)$$

Skewness (*Skew*): The Skewness or distribution spread of the data points of the signal can be calculated as-

$$Skew(x) = \frac{\sum_{i=1}^n (x_i - \mu)^3}{\sigma^3} \quad (0.8)$$

Number of Zero-Crossing (*zCross*): The Number of Zero-Crossing value of all data points during sign change can be calculated as-

$$zCross(x) = \sum_{i=1}^{n-1} \text{sign}(x_i, x_{i+1}) \quad (0.9)$$

$$\text{Where } \text{sign}(a, b) = \begin{cases} 1 & \text{if } (a \times b) < 0 \\ 0 & \text{if } (a \times b) > 0 \end{cases}$$

Positive to Negative Sample Ratio (*PN-Ratio*): The Positive to Negative Sample Ratio or the ratio of the number of positive samples to the number of negative samples of the signal can be calculated as-

$$PN\text{-Ratio}(x) = \frac{\text{len}(ps)}{\text{len}(ns)} \quad (0.10)$$

Where $\text{len}(lst)$ is a function that calculates the length of a list of items lst or counts the total number of items in it. And $ps = \{x_i, x_i \geq 0 : i = 1 \dots n\}$ and $ns = \{x_i, x_i < 0 : i = 1 \dots n\}$.

Positive to Negative Peak Ratio (*PN-PkRatio*): The Positive to Negative Peak Ratio or the ratio of the number of positive peak values to the number of negative peak values of the signal can be calculated as-

$$PN\text{-PkRatio}(x) = \frac{\text{len}(psPk)}{\text{len}(nsPk)} \quad (0.11)$$

Where $\text{len}(lst)$ is a function that calculates the length of a list of items lst or counts the total number of items in it. And $psPk$ = number of positive peaks and $nsPk$ = number of the negative peaks.

Mean Absolute Value (*MeanAbs*): The Mean Absolute Value or the average of the absolute values of all data points can be calculated as-

$$\text{MeanAbs}(x) = \text{Mean}(\{|x_i| : i = 1 \dots n\}) \quad (0.12)$$

Along with the linear features from the time domain, several non-linear features from entropy and complexity measures were calculated. Different entropy measures considered for analysis in this study that measure the quality of meaning information of the signal can be calculated as follows:

Approximate Entropy (*ApEn*): The Approximate Entropy is a technique used to quantify the amount of regularity and the unpredictability of fluctuations over time-series data.

$$\text{ApEn}(x, m, r) = \Phi_1^m(r) - \Phi_1^{m+1}(r) \quad (0.13)$$

Where m is the embedding dimension, r is the similarity criterion or the radius of the neighbourhood. The function Φ_1 is defined as,

$$\Phi_1^m(r) = \frac{1}{n - m + 1} \sum_{i=1}^{n-m+1} \log(C_i^m(r)).$$

The quantity $C_i^m(r)$ is the fraction of patterns of length m that resemble the pattern of the same length that begins at interval i ,

$$C_i^m(r) = \frac{\text{len}(\{x_j, \text{dis}(x_i, x_j) \leq r\})}{n - m + 1}$$

where $\text{dis}(x_i, x_j) = \text{Max}(|u(x_i), u(x_j)|)$, here $\text{dis}()$ is the function that calculates the distance between vectors x_i and x_j ; and u is the corresponding scalar for the respective vector.

Sample Entropy (*SampEn*): Sample Entropy is a modification of approximate entropy used for assessing the complexity of physiological time-series signals. It has two advantages over

approximate entropy: data length independence and a relatively trouble-free implementation.

$$SampEn(x, m, r) = \Phi_1^m(r) - \Phi_2^{m+1}(r) \quad (0.14)$$

Since it is derived from *ApEn*, all other notations are applied in a similar way. The function Φ_1 is as described in *ApEn* and Φ_2 is defined as-

$$\Phi_2^{m+1}(r) = \frac{1}{n-m} \sum_{i=1}^{n-m} \log(C_i^{m+1}(r)).$$

Permutation Entropy (*PermEn*): Permutation Entropy is a measure of complexity based on the probability distribution of the ordinal or permutation pattern or the signal.

$$PermEn(x, m) = - \sum_{i=1}^{m!} P(x_i) \log_2(P(x_i)) \quad (0.15)$$

Here, $m!$ is the factorial of m , and $P(x_i)$ is the probability mass function that provides the probability of a random variable equal to some value. Alternatively, it is the arbitrary discrete probability distribution measured as-

$$P(x_i) = Pr(x = x_i).$$

Singular Value Decomposition Entropy (*SVDEn*): Singular Value Decomposition Entropy is the measure of the dimensionality of the data that indicates if the number of eigenvectors is adequate to explain the data.

$$SVDEn(x, \bar{M}, \tau) = - \sum_{i=1}^{\bar{M}} \bar{\sigma}_i \log_2(\bar{\sigma}_i) \quad (0.16)$$

In this case, τ indicates the time delay or lag between signal points, $\bar{M}$ is the total number of singular values, and $\bar{\sigma}$ is the normalised singular value, and it can be calculated from embedded normalised embedded matrix Y .

$$Y = [y(1), y(2), y(3), \dots, y(n - (m - 1)) \cdot \tau]^T$$

Here, $y(i) = [x_i, x_{i+\tau}, \dots, x_{i-(m-1)\cdot\tau}]$ and Y^T is transpose of the vector Y .

Fuzzy Entropy (*FuzzEn*): Fuzzy Entropy is a similar measurement process of conditional probability as *SampEn* that uses the Fuzzy membership similarity function to match two subsequent signal segments.

$$FuzzEn(x) = \log(\Phi_2^m(r)) - \log(\Phi_2^{m+1}(r)) \quad (0.17)$$

All the parameters of this method are similar to *SampEn*.

Distribution Entropy (*DistEn*): Distribution Entropy is the modified version of *SampEn*, since *SampEn* does not mitigate the problem with its stability and consistency caused by incomplete use of vector-to-vector distance calculation. It is a modified method where histograms were used to estimate the empirical probability density function (ePDF) (1). It can be expressed in terms of Shannon Entropy (*ShanEn*) as follows-

$$DistEn(x, m, M) = \frac{1}{\log_2(M)} \sum_{t=1}^M P_t \log_2(P_t) \quad (0.18)$$

Here, M is the number of bins in the histogram, and P_t is the discrete probability distribution or probability of each t -th bin.

***ShanEn*:** Shannon Entropy provides the level of 'information', or 'uncertainty' on average, that is essential in the corresponding random variable's possible results.

$$ShanEn(x, m, r) = - \sum_{i=1}^n P(x_i) \log(P(x_i)) \quad (0.19)$$

Renyi Entropy (*RenEn*): Like any other entropy method, Renyi Entropy also quantifies the uncertainty or randomness. The *RenEn* of order α , where $\alpha \geq 0$ and $\alpha \neq 1$, is defined as-

$$RenEn(x, \alpha) = \frac{1}{1 - \alpha} \log \left(\sum_{i=1}^n (P(x_i))^\alpha \right) \quad (0.20)$$

Some more complex measurement features are implemented to capture the irregular pattern precisely.

Lempel-Ziv Complexity (*LZComp*): Lempel-Ziv Complexity is used to determine the repetition complexity of signals, in which the signal values are converted to symbols to compute the repeating sub-sequences of symbols (2). For a complexity counter $c(n)$ (calculation procedure is available in (2)) the normalised *LZComp* can be defined as-

$$LZComp(x) = \frac{c(n) \log_2(n)}{n} \quad (0.21)$$

Hjorth Parameters: Hjorth parameters are the statistical features of time-series data. This includes Hjorth mobility and complexity; these can be expressed as below.

Hjorth Mobility (*HjMob*): For the time-series signal x with time t .

$$HjMob(x) = \sqrt{\frac{Var\left(\frac{dx}{dt}\right)}{Var(x)}} \quad (0.22)$$

Here, $\frac{dx}{dt}$ is the first order differentiation of x with respect to time, t .

Hjorth Complexity (*HjComp*):

$$HjComp(x) = \frac{HjMob\left(\frac{dx}{dt}\right)}{HjMob(x)} \quad (0.23)$$

Fisher Information (*FishInfo*): Fisher Information is the way of measuring the amount of information about an unknown parameter θ from x . The simplified form of *FishInfo* can be written as-

$$FishInfo(x, \theta) = \int_{-\infty}^{+\infty} \left(\frac{\partial}{\partial \theta} \iota(x = x_i | \theta) \right)^2 P(x_i) dx \quad (0.24)$$

Here $\iota(x = x_i | \theta)$ is the log-likelihood function with respect to θ .

Fractal Dimensions: Fractal dimension is the measure for characterising fractal patterns by quantifying their complexity in terms of the ratio of change in detail to the change in scale. Some of the most popular fractal measurements are presented below.

Petrosian Fractal Dimension (*PetFd*):

$$PetFd(x) = \frac{\log(n)}{\log(n) + \log\left(\frac{n}{n + 0.4n_\Delta}\right)} \quad (0.25)$$

Where n_Δ is the number of sign changes in the signal derivative.

$$n_\Delta = \sum_{i=1}^{n-2} \left| \frac{z_{i+1} - z_i}{2} \right| \text{ and } z_i = \begin{cases} 1 & \text{if } x_i > Mean(x) \\ -1 & \text{if } x_i \leq Mean(x) \end{cases}$$

Katz Fractal Dimension (*KatzFd*):

$$KatzFd(x) = \frac{\log(n)}{\log(n) + \log\left(\frac{d}{n}\right)} \quad (0.26)$$

Where d is the Euclidean distance between the first and the furthest data points.

HiguchiFractal Dimension (*HigFd*): Unlike other fractal dimensions, HiguchiFractal Dimension uses a transformed signal. If the reconstructed signal of x be-

$$x_i^d = x_i, x_{(i+d)}, x_{(i+2d)}, \dots, x_{(i+\lceil \frac{n-i}{d} \rceil)}$$

Here, $\lceil \frac{n-i}{d} \rceil$ represents the integer part of $\frac{n-i}{d}$.
For every x_i^d the average length-

$$L_i(d) = \frac{\sum_{j=1}^{\lceil \frac{n-i}{d} \rceil} \text{abs}(x_{(i+jd)} - x_{(i+(j-1) \cdot d)}) \cdot (n-1)}{\lceil \frac{n-i}{d} \rceil \cdot d}$$

Therefore, the total average length of the signal-

$$L(d) = \sum_{j=1}^d L_j(d)$$

Finally, by logarithmic transform-

$$\ln(L(d)) \propto \text{HigFd}(x) \cdot \ln\left(\frac{1}{d}\right) \quad (0.27)$$

Detrended Fluctuation (*DetFluc*): In chaotic systems, Detrended Fluctuation is used to determine the statistical self-affinity of the signal x . To do that, the bounded signal x is transformed to unbounded X as follows-

$$X = \sum_{i=1}^n (x_i - \mu)$$

X is also called the cumulative sum or profile. X is divided into time windows of length n samples each, and a local least-squares straight-line fit (the local trend) is calculated by minimising the squared errors within each time window. Let Y indicate the resulting piece-wise sequence of straight-line fits. Then the fluctuation is calculated as follows-

$$\text{DetFluc}(x) = \sqrt{\frac{1}{n} \sum_{i=1}^n (X - Y)^2} \quad (0.28)$$

Sample Entropy Profiling: Additionally, a recent method of Sample Entropy profiling (3) is used to estimate secondary features such as Total SampEn (*TotalSampEn*), Average SampEn (*AvgSampEn*), Maximum SampEn (*MaxSampEn*), Minimum SampEn (*MinSampEn*), Median SampEn (*MedSampEn*), Standard Deviation SampEn (*StdSampEn*), Variance SampEn (*VarSampEn*), Kurtosis SampEn (*KurSampEn*) and Skewness SampEn (*SkewSampEn*) from the profiling information.

In entropy profiling for *SampEn*, instead of choosing or guessing the value of r arbitrarily, a range of values ($1 \leq q \leq n_{bin}$) is generated to generate a list or profile of *SampEn*. Here q is the range, and n_{bin} is the number of elements in the range of cumulative distribution matrix *cdf* of order m . This approach proved promising in many studies without guessing the value for r . Then the profiled *SampEn* can be written as-

$$\text{Prof} - \text{SampEn}(x, m, q) = \ln \frac{\Phi_3^m(q)}{\Phi_3^{m+1}(q)} \quad (0.29)$$

The function Φ_3 can be expressed as,

$$\Phi_3^m(q) = \frac{1}{n-m} \sum_{i=1}^{n-m} \text{cdf}_{iq}^m.$$

The cumulative distribution matrix for m dimension can be generated as-

$$cdf^m = \begin{bmatrix} cdf_1^m(1) & cdf_1^m(2) & \cdots & cdf_1^m(n_{bin}) \\ cdf_2^m(1) & cdf_2^m(2) & \cdots & cdf_2^m(n_{bin}) \\ \cdots & \cdots & \cdots & \cdots \\ cdf_{n-m}^m(1) & cdf_{n-m}^m(2) & \cdots & cdf_{n-m}^m(n_{bin}) \end{bmatrix}$$

And the cumulative distribution function cdf_i^m for distance vector d_i^m can be calculated as-
 $cdf_i^m(q) = p(d_i^m \leq range(q))$; for $1 \leq q \leq n_{bin}$

Where the distance vector can be expressed in terms of adjacent distances,
 $d_i^m = dis(x_i, x_j) : 1 \leq j \leq (mm), j \neq i$

As mentioned, the secondary measures from the profiled *SampEn* are calculated using the fundamental statistical methods described earlier in this subsection. But this time, instead of sending the original signal x of length n , the profile *Prof-SampEn* having the length n_{bin} is sent itself. For instance, the *TotalSampEn* is calculated as-

$$TotalSampEn(x, m, q, n_{bin}) = \sum_{i=1}^{n_{bin}} Prof - SampEn(x, m, q) \quad (0.30)$$

Similarly, all other statistical secondary features (*AvgSampEn*, *MaxSampEn*, *MinSampEn*, *MedSampEn*, *StdSampEn*, *VarSampEn*, *KurSampEn*, *SkewSampEn*) are calculated from the profile of *SampEn*.

Frequency-domain Features

Linear statistical features such as *Max*, *Min*, *Mean*, *Med*, *Std*, *Var*, *Kur* and *Skew* (of the signal) and nonlinear features like Spectral Entropy (*SpecEn*) and BandPower (*BandPw*) are estimated after transforming the given signal x to its frequency domain equivalent signal Fx using Fast Fourier Transform (FFT). Fourier Transform (FT) converts signals from the time/space domain to the frequency domain. Discrete Fourier Transform (DFT) is an FT that works on a finite set of data. The discrete Fourier transform transforms a sequence of n complex numbers $x = x_i : i = 1, 2, 3, \dots, n$ into another sequence of complex numbers, $X = X_i : i = 1, 2, 3, \dots, n$ which is defined by,

$$X_i = \sum_{k=1}^n \left(x_k \cdot e^{-j \frac{2\pi}{n} ik} \right) \quad (0.31)$$

Here j denotes the imaginary part of the complex number. This can also be written after simplification using Euler's formula as-

$$X_i = \sum_{k=1}^n \left(x_k \cdot \left[\cos \frac{2\pi}{n} ik - j \cdot \sin \frac{2\pi}{n} ik \right] \right)$$

For FFT, the data are divided into even and odd indexes. This process makes the same calculation as DFT faster and uses the same DFT matrix. The equation of FFT can be written from DFT as-

$$X_i = X_k(even) + X_k(odd) \cdot e^{-j \frac{2\pi}{n} ik} \quad (0.32)$$

In the frequency domain, features are extracted from (1) the entire spectrum and (2) functional electroencephalography (EEG) bands such as alpha, beta, delta, theta and gamma, each having a different frequency range. The corresponding frequency bands (frequency ranges) Delta (0.4–4 Hz), Theta (4–8 Hz), Alpha (8–12 Hz), Beta (12–30 Hz), Gamma (30 Hz or above) are used to filter the band information. Finally, these band-based signals are used to extract the above-mentioned features.

Band Power: Band power is the integral approximation of the power spectrum using a parabola (Simpson's rule). The segment's absolute *BandPw* and separate frequency bands are calculated.

Appendix-B

Additional Figures

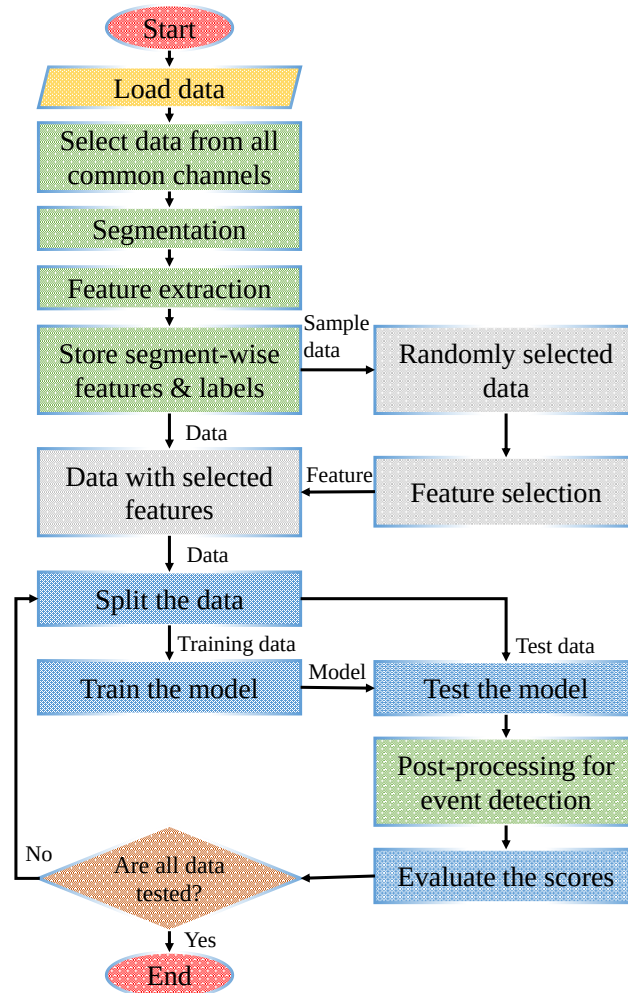


Figure 1: The flowchart explaining the methods followed for the analysis of the data. First, the data are loaded from the dataset. After analysing the channels that are common to all the subjects are filtered from those that are not. Then the data are segments with a 5-second window to extract different features from the time and frequency domain. The features and corresponding labels for the segments are stored for further use (so that the same work need not be done again). Some random data are used for feature selection, this is repeated five times with different channels to get the average feature scores. The next step is to finally select the data with selected features. Based on the experiment type: 5-fold or subject-wise, the following steps are done repeatedly until all data are processed. Data are split into corresponding training and test sets. The machine learning (ML) model is then trained with the training data. The next step is to test the trained model with the test data that are already separated during splitting. A post-processing model is employed to detect seizure events from the classified segments. Lastly, the event detection scores are computed and stored, until all the data are tested and evaluated.

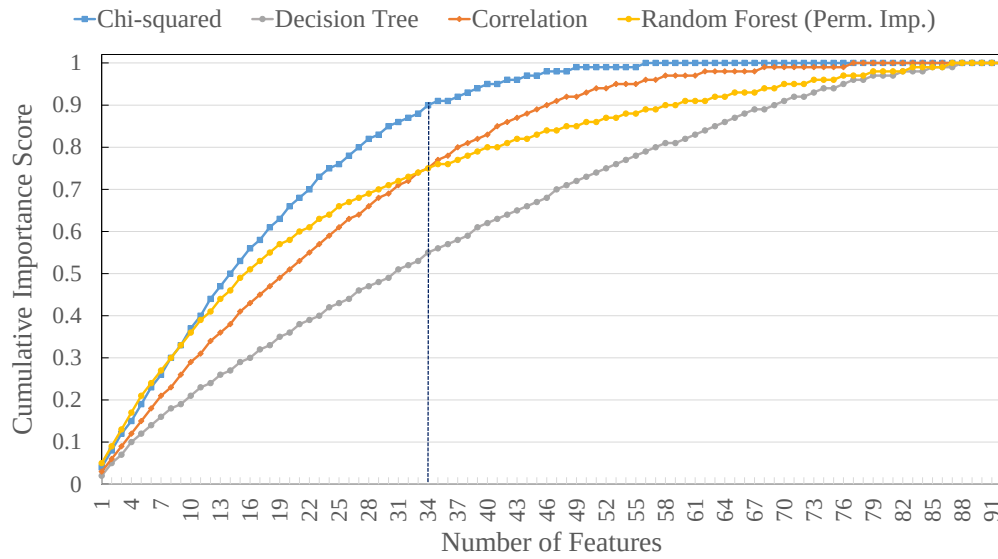


Figure 2: Cumulative features importance score on a scale of 1.0 with respect to the number of features. The graph shows a total of 34 features out of 92 are found more promising in seizure classification for Chi-squared-based features. All other feature selection methods, such as Correlation-based, Decision tree, and Random forest with Permutation feature selection (Perm. Imp.) criteria, are shown alongside this graph. The target cumulative importance of 0.9 (90%) is marked for more visual clarity.

Appendix-C

Additional Data Tables

This section contains additional data tables to demonstrate and present the dataset and data, channel groups and channels, features, and results.

Table 1: This table represents different channel-groups. Each group is enlisted with the number of channels, channel names, and the number of records associated with the channel group.

Group	No. of channels	Channel combination	No. of records
I	18	FP1-F7, F7-T7, T7-P7, P7-O1, FP1-F3, F3-C3, C3-P3, P3-O1, FZ-CZ, CZ-PZ, FP2-F4, F4-C4, C4-P4, P4-O2, FP2-F8, F8-T8, T8-P8, P8-O2	27
II	22	FP1-F7, F7-T7, T7-P7, P7-O1, FP1-F3, F3-C3, C3-P3, P3-O1, FP2-F4, F4-C4, C4-P4, P4-O2, FP2-F8, F8-T8, T8-P8, P8-O2, FZ-CZ, CZ-PZ, P7-T7, T7-FT9, FT9-FT10, FT10-T8	616
III	24	F7-CS2, T7-CS2, P7-CS2, FP1-CS2, F3-CS2, C3-CS2, P3-CS2, O1-CS2, FZ-CS2, CZ-CS2, PZ-CS2, FP2-CS2, F4-CS2, C4-CS2, P4-CS2, O2-CS2, F8-CS2, T8-CS2, P8-CS2, C2-CS2, C6-CS2, CP2-CS2, CP4-CS2, CP6-CS2	1
IV	26	FP1-F7, F7-T7, T7-P7, P7-O1, FP1-F3, F3-C3, C3-P3, P3-O1, FZ-CZ, CZ-PZ, FP2-F4, F4-C4, C4-P4, P4-O2, FP2-F8, F8-T8, T8-P8, P8-O2, FC1-Ref, FC2-Ref, FC5-Ref, FC6-Ref, CP1-Ref, CP2-Ref, CP5-Ref, CP6-Ref	1
V	31	FP1-F7, F7-T7, T7-P7, P7-O1, FP1-F3, F3-C3, C3-P3, P3-O1, FZ-CZ, CZ-PZ, PZ-OZ, FP2-F4, F4-C4, C4-P4, P4-O2, FP2-F8, F8-T8, T8-P8, P8-O2, P7-T7, T7-FT9, FT9-FT10, FT10-T8, FC1-Ref, FC2-Ref, FC5-Ref, FC6-Ref, CP1-Ref, CP2-Ref, CP5-Ref, CP6-Ref	39

Table 2: Data description: count of total records, records with seizure events, and total seizure episodes for each subject available in the Children’s Hospital Boston-Massachusetts Institute of Technology (CHB-MIT) dataset. The minimum and maximum duration of the events, along with the total duration of seizure data, are provided in the right columns of this table.

Subject	No. Records	No. Records with Seizures	No. Seizure Events	Seizure Duration (Min-Max)	Total Seizure Duration
01	42	7	7	28–102	449
02	36	3	3	10–83	175
03	38	7	7	48–70	409
04	42	3	4	50–117	382
05	39	5	5	97–121	563
06	18	7	10	13–21	147
07	19	3	3	87–144	328
08	20	5	5	135–265	924
09	19	3	4	63–80	280
10	25	7	7	36–90	454
11	35	3	3	23–753	809
12	21	13	40	7–90	1475
13	11	8	12	18–71	547
14	26	7	8	15–42	117
15	39	14	20	31–205	2012
16	17	6	10	7–15	94
17	20	3	3	89–116	296
18	35	6	6	31–69	323
19	29	3	3	78–82	239
20	29	6	8	30–50	302
21	33	4	4	13–82	203
22	31	3	3	59–75	207
23	9	3	7	21–114	431
24	22	12	16	17–71	527
Total	686	141	198	7–753	11693

All duration are in seconds and are approximate.

Table 3: List of features extracted from scalp electroencephalography segments. This list represents different types of features extracted from different domains (e.g. time and frequency domain) along with the number of features extracted from that particular category. The feature names are provided in their abbreviated form, the elaboration is provided in the description section of the features.

Feature Details			No. of features
Domain	Type	Feature name	
Time	Statistical	Maximum, Minimum, Mean, Median, Standard Deviation, Variance, Kurtosis, Skewness, Number of Zero-Crossing, Positive to Negative Sample Ratio, Positive to Negative Peak Ratio, Mean Absolute Value	12
	Entropy	Approximate Entropy, Sample Entropy, Permutation Entropy, Singular Value Decomposition Entropy, Fuzzy Entropy, Distribution Entropy, Shannon Entropy, Renyi Entropy	8
	Complexity and Fractal Dimension	Lempel-Ziv Complexity, Hjorth Mobility, Hjorth Complexity, Fisher Information, Petrosian Fractal Dimension, Katz Fractal Dimension, HiguchiFractal Dimension, Detrended Fluctuation	8
	Sample Entropy Profile	Total SampEn, Average SampEn, Maximum SampEn, Minimum SampEn, Median SampEn, Standard Deviation SampEn, Variance SampEn, Kurtosis SampEn, Skewness SampEn	9
Frequency	Statistical	Maximum, Minimum, Mean, Median, Standard Deviation, Variance, Kurtosis, Skewness	8
	Band-wise Statistical	Maximum, Minimum, Mean, Median, Standard Deviation, Variance, Kurtosis, Skewness (from the band: alpha, beta, delta, theta and gamma)	40
	Entropy	Spectral Entropy	1
	Power	BandPower	1
	Band-wise Power	BandPower (from the band: alpha, beta, delta, theta and gamma)	5
Total Features			92

Table 4: List of features selection methods that are used in this study along with the number of features selected from each channel using that method. The number of features is selected based on 90% of the cumulative scaled feature importance values. These features include the top most important features that provide maximum classification performance.

Method	Number of features selected
Chi-squared	34
Correlation (with target)	46
DT	69
RF (perm. imp.)	58

Table 5: List of features selected from all the extracted features from the scalp electroencephalography segments. This list represents different types of features extracted from different domains (e.g. time and frequency domain) along with the number of features extracted from that particular category. The feature names are provided in their abbreviated form, the elaboration is provided in the description section of the features. The note provided in the table denotes the short names used in the table for different frequency sub-bands.

Feature Details			No. of features
Domain	Type	Feature name	
Time	Statistical	Maximum, Standard Deviation, Variance, Mean Absolute Value	4
	Entropy	Renyi Entropy	1
	Complexity and Fractal Dimension	Lempel-Ziv Complexity	1
	Sample Entropy Profile	Total SampEn	1
Frequency	Statistical	Maximum, Mean, Median, Standard Deviation, Variance	5
	Band-wise Statistical	Maximum(abg), Minimum(abg), Mean(abtg), Median(abg), Standard Deviation(abg), Variance(abg)	19
	Entropy	×	0
	Power	BandPower	1
	Band-wise Power	BandPower(dt)	2
Total Features			34
The letters a,b,d,t, and g for all 5 frequency sub-bands (alpha, beta, delta, theta, and gamma)			

Table 6: The random forest classifier is chosen for the analysis of this study. This table shows the performance of different models for the same data and the same data splitting. Based on different performance metrics random forest is found to be consistently doing better in terms of many metrics. Mainly based on the Accuracy (Acc) and F1 scores, RF model was chosen for experimentation.

Classifier Model	Acc (%)	AUC (%)	Sen (%)	Pre (%)	F1
Random Forest	99.85	96.04	34.43	93.20	50.19
Extra Tree	99.84	96.57	30.36	96.56	46.14
Ada Boost	99.79	98.21	21.34	57.55	31.04
K-Nearest Neighbours	99.78	64.73	07.71	60.08	13.63
Decision Tree	99.73	72.17	44.48	39.89	42.02
Support Vector Machine	97.45	0.00	10.26	01.17	01.40
Naïve Bayes	94.82	91.83	51.98	02.24	04.30
Acc=Accuracy, AUC=Area under the ROC (receiver operating characteristic) Curve, Sen=Sensitivity, Pre=Precision, F1=F1 Score.					

Extended result table

This table represents the extended results presented in Table 3 of the main article with all other performance measurement metrics. The first section of the table contains the epileptic seizure segment-based results, and the last one is for event-based results. In addition to Sensitivity (*Sen*) and false detection rate (*FDR*) scores, the include Accuracy (*Acc*), Specificity (*Spe*), Precision (*Prec*) and F1 Score (*F1S*). The first row demonstrates the result of the experiment where the 5-fold randomly selected samples were used to test the performance. On the other hand, the last two rows in this section show the segment-based results for subject-wise (cross-subject) analysis. Two rows of the second section of the table list the event-based performance scores for the same results stated in the last two rows of the above section of the table, but this result is calculated after doing the post-processing for event detection. The segment-based results show the mean $\pm$ SD (standard deviation) results for all the scores for 5 different runs of the same experiment. On the other hand, the event-based results for one experiment are taken from one of the segment-based results.

Table 7: Additional summary of the performance scores extending Table-3 results for this study for the event-based and segment-based results. This table includes all the relevant available performance metrics such as *Acc*, Recall (*Rec*)/*Sen*, *Spe*, *Prec*, *F1S* and *FDR* for the corresponding study. 'NR' in this table indicates that the result is not reported or available. The event-based results are our main target to observe and thus are added to the main article and comparison table. The segment-based results are presented in mean $\pm$ SD (standard deviation) format for 5 runs.

Study	<i>Acc</i> (%)	<i>Sen</i> (%)	<i>Spe</i> (%)	<i>Prec</i> (%)	<i>F1S</i> (%)	<i>FDR</i> (/hr)
Segment-based results						
Random sample 5-fold	94.08 $\pm$ 0.18	91.88 $\pm$ 0.43	94.09 $\pm$ 0.18	4.37 $\pm$ 0.14	8.33 $\pm$ 0.26	NR
Subject-wise 5-fold	85.06 $\pm$ 3.35	64.24 $\pm$ 2.57	85.12 $\pm$ 3.35	2.97 $\pm$ 1.57	5.46 $\pm$ 2.65	NR
Subject-wise Leave-one-out (LOO)	85.96 $\pm$ 2.80	66.75 $\pm$ 1.43	86.09 $\pm$ 2.09	5.84 $\pm$ 1.33	9.39 $\pm$ 2.02	NR
Event-based results						
Subject-wise 5-fold (proposed)	47.95	72.63	28.56	41.23	51.49	5.32
Subject-wise LOO (proposed)	49.59	75.34	29.26	43.72	54.33	4.79

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